llmXive Automated Review of DragMesh-2: Physically Plausible Dexterous Hand-Object Interaction with Articulated Objects

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and their alignment with the provided data and citations.

1. Inconsistency in Benchmark Scope (Section 4 vs. Tables 1 & 2): The Introduction and Section 4 (Experiments) state the evaluation is conducted on a benchmark of "7 GAPartNet objects spanning three categories (Dishwasher, StorageFurniture, Microwave)". However, Table 1 (Dataset) and Table 2 (Results) explicitly list **seven** distinct categories: StorageFurniture, TrashCan, Dishwasher, Refrigerator, Oven, Microwave, and TableObject. The text's claim of "three categories" is factually incorrect based on the provided tables. While the *results* might be aggregated or focused on a subset, the text explicitly claims the benchmark *spans* three categories, which contradicts the data presented in the tables showing seven. This requires a correction to accurately reflect the experimental setup.

2. Nuance in Parallel-Jaw Primitive Performance: The text claims the GT-part-pose parallel-jaw primitive is "damping-invariant" and succeeds on "only one object (0.14 mean)". While the average (0.14) and the invariance (1.00 at ×1, ×2, ×4 for the single success) are mathematically correct, the phrasing "succeeds on only one object" combined with "damping-invariant" could be misinterpreted as a uniform failure across all objects. The data shows it succeeds perfectly (1.00) on object 12583 (Dishwasher) regardless of damping.

The claim is technically accurate but lacks the nuance that the primitive is highly effective for specific geometries (like the dishwasher door) but fails completely for others. This is a minor writing issue but impacts the precision of the scientific claim regarding the primitive's generalizability.

3. Interpretation of Ablation Results (Section 4, "Ablation" paragraph): The text claims: "the physical signals contribute more under nominal damping while the temporal encoder helps more under stochastic mid-damping." - **Nominal (×1):** "w/o PICA (GLA only)" = 0.65; "w/o GLA (PICA only)" = 0.75. Here, removing the temporal encoder (keeping physical signals) yields a *higher* score (0.75) than removing physical signals (0.65). This supports the claim that physical signals are crucial at nominal damping. - **Stochastic Mid-Damping (×2):** "w/o PICA (GLA only)" = 0.64 (stochastic); "w/o GLA (PICA only)" = 0.57 (stochastic). Here, the model *with* the temporal encoder (GLA only) performs slightly better (0.64) than the one without (0.57). However, the text states the temporal encoder helps *more* under stochastic mid-damping. The difference (0.64 vs 0.57) is marginal (0.07), whereas the drop in deterministic performance for "w/o GLA" at ×2 is significant (0.71 vs 0.56). The claim that the temporal encoder is the primary driver for stochastic mid-damping robustness is weakly supported by the data, as the "w/o PICA" (GLA only) model actually has the highest stochastic score at ×2, but the "w/o GLA" (PICA only) model has a much lower deterministic score. The text's interpretation of "helps more" is ambiguous and potentially overstated given the small margin in stochastic performance and the large drop in deterministic performance when removing the encoder. The claim should be tempered or the data re-examined to ensure the conclusion matches the magnitude of the effect.

4. Citation Verification: The citations (e.g., 'okami2024', 'bao2023dexart') are standard placeholders in the provided LaTeX source. Without the actual '.bib' file content, I cannot verify if the *content* of the cited papers matches the claims (e.g., "Dexterous hands provide more compliant multi-finger contact patterns"). However, the claims themselves are general knowledge in the field and the citations are plausible. No obvious factual errors in the *attribution* of specific results to these papers were found in the text provided, assuming the bibliography is correct.

Conclusion: The paper contains a clear factual error regarding t

- [writing] Section 4 claims the benchmark spans 'three categories', but Tables 1 and 2 list seven (including TrashCan, Oven). Correct the text to match the data.
- [writing] The claim that the parallel-jaw primitive is 'damping-invariant' with '0.14 mean' obscures that it succeeds perfectly (1.00) on object 12583. Clarify this nuance.
- [science] The ablation claim that the temporal encoder helps 'more' under stochastic mid-damping is weakly supported; the stochastic gain (0.64 vs 0.57) is marginal compared to deterministic drops.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
The figure displays a single render of a robotic hand grasping a cabinet, but the caption misleadingly claims to show three trajectories. Additionally, the caption contains a capitalization error and includes a raw filename artifact.
Figure 2
The figure fails to support its caption's claim of showing diversity, as it presents only a single unlabeled object without any visual indicators of category or joint-type variation.
Figure 3
Figure 3 is a clear hardware snapshot showing a robotic arm and storage unit, consistent with its caption describing it as a hardware stage snapshot. As a qualitative visual, it lacks axes or legends, which is appropriate for this figure type.
Figure 4
Figure 4 provides a clear and comprehensive visual overview of the DragMesh-2 architecture, effectively breaking down the system into its core components: Temporal Awareness, Physical Causality, RL Loop, and Damping Randomization. The diagram is well-structured with distinct color coding and clear labeling of inputs, outputs, and internal processes, making the complex pipeline easy to follow.
Figure 5
The figure is visually clear with a comprehensive legend, but the caption contains a formatting typo and cites specific success rate values that contradict the visual data shown in the bar chart.
Figure 6
The figure is incomplete; the caption describes a two-panel layout (hardware top, simulation bottom), but the image only displays a single hardware photograph with no simulated rollout visible.

- [writing] Figure 1: The caption claims to show 'Three generated object trajectories,' but the image displays only a single snapshot of a single trajectory instance, failing to provide the promised comparative view.
- [writing] Figure 1: The caption contains a capitalization error ('Three' should be lowercase) and includes a raw filename ('[sim_46440_approach.png]') that should be removed.
- [science] Figure 2: The caption claims to show 'category and joint-type diversity' in terminal-stage renders, but the image displays a single static object (a cabinet) with no labels, annotations, or comparative views to demonstrate this diversity.
- [writing] Figure 2: The image lacks a figure label (e.g., 'Figure 2') and contains no text, scale bars, or legends to identify the object category or joint types mentioned in the caption.
- [writing] Figure 5: The caption contains a formatting error 'mode$$damping' and in-

cludes specific numerical claims (e.g., '0.56 deterministic at 4') that contradict the visual data in the chart, where the PICA bar at x4 in panel A is clearly below 0.50.

- [science] Figure 5: The caption claims the 'GT Parallel-Jaw' primitive is deterministic and identical across modes, yet the legend includes it in both panels; while the bars appear similar, the caption's specific numerical examples for PICA are factually inconsistent with the plotted heights.
- [science] Figure 6: The caption states the figure contains a 'Top' hardware example and a 'Bottom' simulated rollout, but the provided image is a single photograph with no bottom panel or simulated data shown.
- [writing] Figure 6: The caption describes a two-part figure (Top/Bottom), but the rendered image does not visually distinguish or separate these components, making the description impossible to verify.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript exhibits a high density of specialized jargon and unexplained acronyms that may hinder accessibility for non-specialist readers in robotics or general AI.

First, the acronym **GLA** (Gated Linear Attention) is introduced in Section 3.2 (Eq. 3) and Appendix 3.3.3 without a prior definition. The text assumes the reader knows this specific attention mechanism variant. It should be spelled out at first use (e.g., "a Gated Linear Attention (GLA) encoder").

Second, the term **loco-manipulation** appears in the Abstract and Section 1. While standard in specific robotics sub-fields, it is not universally understood. Replacing it with "whole-body manipulation" or "mobile manipulation" would improve clarity for a broader audience.

Third, the proposed method **PICA** is introduced in the Abstract without its full expansion. While the Introduction defines it as "Physically Informed Contact-Aware," the Abstract should be self-contained. The full name should appear in the Abstract upon first mention.

Fourth, the phrase **action-boundary regularization** in Section 3.2 is jargon-heavy. It describes a penalty for actions near joint limits. Rewriting this as "regularization to prevent actions from hitting joint limits" or similar plain English would make the mechanism clearer to readers unfamiliar with specific RL constraint formulations.

Finally, the acronym **OOD** (Out-of-Distribution) is used in Section 4 regarding "OOD contact-load shift." While common in machine learning, it should be explicitly defined as "out-of-distribution" at its first occurrence to ensure clarity for readers from other domains.

These changes are minor but necessary to ensure the paper is accessible to the broader robotics and AI community beyond the immediate sub-specialty.

- [writing] Define the acronym 'GLA' (Gated Linear Attention) at its first occurrence in Section 3.2 (Eq. 3) and Appendix 3.3.3. Currently, it appears without definition, assuming reader familiarity with specific attention variants."
- [writing] Replace the term 'loco-manipulation' with 'whole-body manipulation' or 'mobile manipulation' in the Abstract and Section 1. 'Loco-manipulation' is a niche robotics term that may exclude non-specialist readers."
- [writing] Define 'PICA' (Physically Informed Contact-Aware) immediately upon its first introduction in the Abstract. While defined in the Introduction, the Abstract should be self-contained for non-specialist readers."
- [writing] Replace the phrase 'action-boundary regularization' with 'penalizing actions near joint limits' or similar plain language in Section 3.2. The current phrasing is dense and assumes knowledge of specific RL regularization techniques."
- [writing] Clarify the term 'OOD' (Out-of-Distribution) at its first use in Section 4. While common in ML, it should be explicitly defined for a broader robotics audience in the context of 'contact-load shift'."

paper_reviewer_logical_consistency

Verdict: minor_revision
The logical consistency of the paper is generally strong, particularly in the formulation of the contact-driven task and the motivation for the PICA mechanism. The central claim—that nominal success metrics are insufficient for articulated object manipulation due to contact-load sensitivity—is well-supported by the experimental data in Table 1 and Table 3. The ablation study (Table 2) logically isolates the contributions of the temporal encoder and physical signals, supporting the conclusion that both are necessary for robustness.

However, there are two areas where the causal links between the proposed mechanisms and the observed outcomes require tighter logical justification:

1. **Auxiliary Target Definition (Section 4.2, Eq. 4):** The paper defines the auxiliary target y_t to include a binary detachment risk indicator alongside continuous metrics. The text argues that these targets allow the policy to learn "contact-conditioned interaction" and infer "contact impedance." Logically, a binary failure flag (detachment) is a discrete event that occurs *after* contact is lost, whereas "impedance" and "stable pulling" are continuous states *during* contact. While the continuous distance term serves as a valid proxy for contact maintenance, the inclusion of the binary detachment flag as a predictor for *stable* interaction is logically tenuous. The binary flag indicates a failure mode, not a gradient for stable control. The authors should clarify whether the binary term is intended to penalize the *approach* to failure or if it is merely a diagnostic. If the latter, its inclusion in the auxiliary loss for learning "contact response" needs a stronger causal explanation, as predicting a binary failure state does not inherently teach the policy how to maintain contact under varying loads.

2. **Causal Mechanism of Saturation (Section 5, ”Nominal success masks saturation collapse”):** The paper asserts that optimizing for task progress alone causes policies to ”overfit nominal dynamics” and ”rely on dynamics shortcuts,” leading to action saturation (high ‘clip099‘) and subsequent OOD failure. The data in Table 3 supports the correlation (higher epochs $\to$ higher saturation $\to$ lower OOD success). However, the logical chain connecting the *reward function* to this specific failure mode is slightly opaque. The reward function (Eq. 5) includes a task progress term and an action energy penalty. The paper implies that the policy finds a ”shortcut” by saturating actions to force movement, suggesting the energy penalty is insufficient to counteract the progress reward under high damping. To fully support the claim that PICA's specific regularizers are the logical solution, the text should explicitly explain *why* the standard PPO optimization converges to this saturated state (e.g., is the gradient of the progress term steeper than the energy penalty in high-damping regimes?). Without this mechanistic explanation, the claim that the failure is a direct result of ”task-progress-only optimization” remains a strong correlation rather than a fully derived causal argument.

Addressing these points will strengthen the logical rigor of the paper's central contributions.

- [science] In Section 4.2 (Eq. 4), the auxiliary target includes a binary detachment flag. The text claims this helps infer 'contact impedance' for stable pulling. Logically, a binary failure flag indicates a state *after* contact loss, not a continuous proxy for impedance *during* contact. Clarify the distinct causal role of this binary term versus the continuous distance term in learning stable interaction.
- [science] Section 5 claims optimizing task progress causes action saturation and OOD failure. While Table 3 shows the correlation, the causal mechanism is unclear: does the reward function explicitly incentivize saturation, or is it a side effect? Explicitly link the reward terms (Eq. 5) to the saturation behavior to justify why PICA's specific regularizers are the necessary logical fix.

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several claims that slightly overreach the empirical evidence provided, particularly regarding the scope of the released dataset and the generalizability of the robustness claims.

First, the Abstract and Conclusion (lines 24-26, 338-340) claim the work provides a ”pure-geometry dexterous interaction resource to support future loco-manipulation.” This is an overstatement. The dataset described in Section 3.3 and Table 1 consists of 277 hand-object trajectories for 7 specific GAPartNet objects. It contains no locomotion data, no whole-body balance information, and no ground-contact dynamics. While the authors correctly note in the Limitations (lines 356-360) that the dataset *could* serve as a prior for whole-body control, presenting it as a resource *for* loco-manipulation implies a completeness it does not possess. The text should be tempered to state the dataset provides a ”motion-scale prior” or ”upper-body initialization” for such future work, rather than a direct resource.

Second, the claim in the Introduction (lines 58-60) that PICA improves robustness ”without requiring additional force sensing” is technically true but risks over-claiming the method’s capability. The Limitations section (lines 348-352) explicitly admits that the lack of force/tactile feedback is the primary reason for failure under strong damping (x4). The current phrasing suggests the method has solved the robustness problem without sensors, whereas the data shows it merely mitigates it better than baselines within a specific, limited regime. The claim should be qualified to ”improves robustness *relative to baselines* without additional force sensing” or ”extends the robustness envelope of kinematic-only policies.”

Finally, the Abstract (line 23) states the method maintains ”high task success” across damping conditions. Table 1 shows a drop from 89% to 56% success at x4 damping. While 56% is the best among baselines, characterizing a 44% failure rate under OOD conditions as ”high task success” is an over-interpretation of the results. It is more accurate to claim the method ”maintains superior relative robustness” or ”retains non-trivial success” under strong contact-load shifts. These adjustments will align the paper’s narrative more strictly with the quantitative evidence.

- [writing] The abstract claims the dataset supports ’future loco-manipulation,’ but Table 1 shows only 277 hand-centric trajectories with no whole-body data. Rephrase to state it provides a ’motion-scale prior’ for such tasks, not a direct resource.
- [writing] The introduction claims robustness ’without force sensing,’ yet limitations admit this causes failure at high damping. Qualify the claim to ’improves robustness relative to baselines’ to avoid implying the sensor-less problem is solved.
- [writing] The abstract describes 56% success at x4 damping as ’high task success.’ This overstates the result. Change to ’maintains superior relative robustness’ or ’retains non-trivial success’ to accurately reflect the 44% failure rate.

paper_reviewer_safety_ethics

Verdict: accept

The manuscript presents a simulation-based reinforcement learning framework (DragMesh-2) for dexterous hand-object interaction with articulated objects. From a safety and ethics perspective, the work is low-risk. The research is conducted entirely within a simulated environment (Isaac Gym/SAPIEN implied by context and standard practice in the field), utilizing the GAPartNet dataset, which consists of synthetic 3D models. There is no collection of human data, no deployment on physical hardware in uncontrolled environments, and no involvement of human subjects or animals; therefore, IRB or IACUC approval is not applicable, and no consent procedures are required.

The potential for dual-use or physical harm is minimal. The system controls a virtual

51-DoF hand to manipulate virtual objects (drawers, doors) in a physics simulator. While the underlying technology (dexterous manipulation) could theoretically be applied to real-world robotics, the paper explicitly states in Section 5 (Limitations) and Figure 1 caption that quantitative evaluations are simulation-only, and hardware images are merely qualitative feasibility illustrations. The paper does not provide instructions for deploying the policy on physical hardware, nor does it claim real-world robustness that would necessitate immediate safety certification.

Data privacy and consent are not concerns as the dataset (GAPartNet) is a public, synthetic benchmark. The authors acknowledge the limitations of their approach, specifically the lack of tactile/force feedback and the risk of action saturation under high damping (Section 5), which are technical limitations rather than safety hazards in the current context. The code and dataset are released via public links (GitHub/Website), which is standard for reproducibility and does not introduce new ethical risks. No conflicts of interest are declared, and the funding sources are standard academic affiliations. The paper adheres to responsible AI research practices by clearly delineating the scope of the work (simulation) and acknowledging the gap to real-world deployment.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The scientific evidence supporting the central claims of DragMesh-2 is generally robust, particularly regarding the ablation of the PICA mechanism and the demonstration of overfitting in standard PPO baselines. The experimental design effectively isolates the contribution of contact-aware signals and temporal encoding. However, several statistical and interpretative issues require clarification before the evidence fully supports the conclusions.

First, the sample size for the main comparison is 20 episodes per cell (Section 5, "Experiments"). For a binary success metric, the standard error is approximately $\sqrt{p(1-p)/20} \approx 0.11$ at $p=0.5$. The paper frequently highlights small absolute differences in success rates (e.g., 0.15 vs. 0.30 at $\times4$ damping in Table 1) as evidence of methodological superiority. Without confidence intervals or statistical significance testing (e.g., McNemar's test or bootstrap CIs), these differences are statistically indistinguishable from noise. The authors should report error bars in Figure 2 and add a statistical analysis section to validate the claimed performance gaps.

Second, the "Traj. tracking" baseline presents a logical anomaly in the evidence. Table 1 shows this open-loop baseline achieving 1.00 success at $\times1$, $\times2$, and $\times4$ damping for most objects, yet the text claims open-loop replay fails under high damping. If the baseline is truly robust to damping changes, the claim that "open-loop replay alone is not OOD-robust" is unsupported by the provided data. The authors must explain why this specific baseline succeeds where learned policies fail, or re-evaluate the baseline's implementation to ensure it does not inadvertently cheat the physics (e.g., by ignoring damping in the trajectory generation).

Third, the claim of "physically plausible" interaction relies heavily on simulation results. While the hardware visualization in Figure 4 is compelling, it is qualitative. The absence of quantitative hardware metrics (e.g., contact force profiles, slip detection rates) limits the ability to verify the physical plausibility claim outside of simulation. The authors should either provide quantitative hardware data or temper the claim to "simulation-validated physical plausibility."

Finally, the training-length study (Table 3) provides strong evidence of the "nominal success masks saturation collapse" phenomenon, which is a key contribution. However, the transition from 0.55 to 0.10 success between 200 and 300 epochs is abrupt. The authors should discuss the sensitivity of this collapse to random seeds to ensure it is a systematic failure mode rather than a stochastic artifact.

- [science] The standard error for success rates (n=20) is ~0.11, yet the paper treats small differences (e.g., 0.15 vs 0.30 at x4 damping) as distinct performance tiers. Report confidence intervals or statistical significance tests for the main comparison in Table 1 and Figure 2 to support claims of superiority.
- [science] The 'Traj. tracking' baseline achieves 1.00 success at x1 and x4 damping (Table 1), which contradicts the claim that open-loop replay fails under high damping. Clarify why this specific baseline is robust to damping changes while learned policies fail, or re-evaluate the baseline's physical validity.
- [science] The hardware results in Figure 4 are qualitative only. To support the claim of 'physically plausible' interaction, provide quantitative metrics (e.g., contact force, slip rate) from the hardware trials or explicitly limit the claim to simulation validity.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The statistical analysis in DragMesh-2 is generally sound in its design, particularly the use of out-of-distribution (OOD) damping conditions to test robustness and the inclusion of diagnostic metrics like action saturation (‘clip099‘) alongside success rates. The authors correctly identify that nominal success can mask failure modes, and the experimental protocol (20 episodes per cell) provides a reasonable sample size for a simulation-based RL benchmark.

However, the presentation of results lacks necessary statistical rigor for a quantitative comparison. In **Table 1** (Main Comparison) and **Table 2** (Ablation), success rates are reported as point estimates (e.g., 0.56 vs 0.50) without confidence intervals (CIs) or standard errors. As noted in the Appendix (Section 5.1, "Limitations"), the standard error for a proportion with $N=20$ is approximately $\sqrt{p(1-p)/20} \approx 0.11$. Consequently, differences of 0.05–0.10 between methods (e.g., PICA vs. Reconfig at $\times4$ damping) are not statistically distinguishable. The text claims PICA "attains the best mean in every damping/mode column," but without error bars or significance testing, these claims are not statistically supported.

Furthermore, the ablation study involves multiple pairwise comparisons across objects and damping levels. The authors do not mention any correction for multiple comparisons (e.g., Bonferroni or False Discovery Rate), increasing the risk of Type I errors when claiming the superiority of the full PICA model over its ablated variants. While the Appendix acknowledges the limitation of small sample sizes, the main text and tables should reflect this uncertainty visually (error bars) or numerically (CIs).

Finally, the independence of the 20 episodes per cell is not explicitly defined. If the same random seed or initial state is used across different damping conditions for the same object, the samples are correlated, which would invalidate standard error calculations. The authors should clarify the randomization protocol for episode initialization and damping sampling in the Appendix to ensure the statistical validity of the reported metrics.

- [science] Report confidence intervals or standard errors for the success rates in Table 1 and Table 2. With N=20 episodes per cell, the standard error is ~0.11; without error bars or CIs, small differences (e.g., 0.56 vs 0.50) are statistically indistinguishable. Appendix Limitations (Sec 5.1) acknowledges this but the main results tables lack the necessary statistical context.
- [science] Clarify the statistical independence of the 20 episodes per cell. If the same random seed or initialization is used across damping conditions for a single object, the samples are not independent, invalidating standard error calculations. Explicitly state the randomization protocol for episode initialization and damping sampling in Appendix Sec 5.1.
- [science] The ablation study (Table 2) compares multiple methods without correcting for multiple comparisons. With 7 objects and 3 damping levels, the number of pairwise tests is high. Report adjusted p-values or use a non-parametric test (e.g., Friedman test) to validate that PICA's superiority over baselines is statistically significant rather than due to chance.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript demonstrates a high level of technical writing with clear logical flow and precise terminology appropriate for the robotics and reinforcement learning community. The structure is well-organized, moving effectively from the problem statement to the proposed method, experiments, and analysis. The use of equations to define the task formulation and the PICA mechanism is clear and mathematically sound.

However, there are minor issues regarding consistency and phrasing that, while not critical, detract slightly from the overall polish. In the Abstract and Introduction, the repeated use of ”hand–object” and ”hand–handle” relies on LaTeX double-hyphens for en-dashes. While standard in source code, the authors should verify the compiled PDF renders these correctly as en-dashes to maintain professional typography. Additionally, in Section 4, the description of damping multipliers uses the phrase ”contact-load shift” three times in close proximity; varying this terminology would improve readability.

There is also a minor ambiguity in the description of the auxiliary targets in Section 3.2 and Appendix A.3.4. The text describes the components of the vector y_t as including ”detachment risk,” while the equation explicitly includes a binary indicator for detachment. Clarifying whether the ”risk” refers to the binary flag or a continuous probability would prevent potential confusion. Finally, the category name ”TableObject” in Table 1 should be cross-checked against the standard GAPartNet taxonomy to ensure consistency with the dataset’s official naming conventions. Addressing these points will further enhance the clarity and professionalism of the paper.

- [writing] In the Abstract and Introduction, the phrase ’hand–object’ and ’hand–handle’ uses double hyphens for en-dashes. While common in LaTeX source, ensure the compiled PDF renders these as proper en-dashes and not double hyphens, as this affects readability and professional appearance.
- [writing] Section 3.2 (Eq. 4) and Appendix A.3.4 (Eq. aux) define the auxiliary target vector y_t. The text describes the components as ’recent object response, maximum palm–handle distance, detachment risk, and tracking stress,’ but the equation includes a binary indicator for detachment. Clarify if the binary indicator is distinct from the ’detachment risk’ description or if the description should explicitly mention the binary nature.
- [writing] In Section 4, the text states ’The damping multipliers ×1, ×2, and ×4 measure nominal performance, mild contact-load shift, and strong out-of-distribution (OOD) contact-load shift, respectively.’ The phrasing ’contact-load shift’ is repeated. Consider varying the vocabulary (e.g., ’increased resistance’ or ’higher damping’) to improve flow and reduce redundancy.
- [writing] The caption for Table 1 (tab:dataset) lists ’TableObject’ with 1 trajectory, but the text in Section 3.3 mentions ’7 GAPartNet categories’ and the table lists 7 rows. Ensure the category name ’TableObject’ is consistent with the GAPartNet taxonomy used in the rest of the paper (e.g., is it ’Table’ or ’TableObject’?) to avoid confusion for readers cross-referencing the dataset.